a) PKCE

b) Hugh grant

c) Implicit grant

d) Authorization code grant

e) Resource owner password credentials grant

11.3 The JWT bearer grant for OAuth2

Authentication with a client secret or service account password is very simple, but suffers from several drawbacks:

- Some features of OAuth2 and OIDC require the AS to be able to access the raw bytes of the client secret, preventing the use of hashing. This increases the risk if the client database is ever compromised as an attacker may be able to recover all the client secrets.
- If communications to the AS are compromised, then an attacker can steal client secrets as they are transmitted. In section 11.4.6 you'll see how to harden access tokens against this possibility, but client secrets are inherently vulnerable to being stolen.
- It can be hard to change a client secret or service account password, especially if it is shared by many servers.